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$$\lim_{x \rightarrow -\sqrt{3}} \frac{x^2 + 2\sqrt{3}x + 3}{x^2 - 3} = \frac{0}{0}$$

We noemen de vergelijking in de teller en noemer respectievelijk  $t(x)$  en  $n(x)$

$t(x)$  is van de vorm  $(a + b)^2$  met:

$$\Rightarrow \begin{cases} a^2 = x^2 & \Rightarrow a = x \\ b^2 = 3 & \Rightarrow b = \sqrt{3} \end{cases}$$

$$t(x) = (x + \sqrt{3})^2 = (x + \sqrt{3})(x + \sqrt{3})$$

$n(x)$  is van de vorm  $a^2 - b^2 = (a + b)(a - b)$  met:

$$\Rightarrow \begin{cases} a^2 = x^2 & \Rightarrow a = x \\ b^2 = 3 & \Rightarrow b = \sqrt{3} \end{cases}$$

$$n(x) = (x + \sqrt{3})(x - \sqrt{3})$$

$$\begin{aligned} & \lim_{x \rightarrow -\sqrt{3}} \frac{t(x)}{n(x)} \\ &= \lim_{x \rightarrow -\sqrt{3}} \frac{\cancel{(x + \sqrt{3})} (x + \sqrt{3})}{\cancel{(x + \sqrt{3})} (x - \sqrt{3})} \\ &= \lim_{x \rightarrow -\sqrt{3}} \frac{x + \sqrt{3}}{x - \sqrt{3}} \\ &= \frac{-\sqrt{3} + \sqrt{3}}{-\sqrt{3} - \sqrt{3}} \\ &= \frac{0}{-2\sqrt{3}} \\ &= 0 \end{aligned}$$

## References